

COSUP

COMMUNITY ORIENTED SUBSTANCE USE PROGRAMME

COSUP

Community Oriented Substance Use Programme

TASK 2 — REQUIREMENT ANALYSIS AND SYSTEM DESIGN

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Module: XISD5319 — Work Integrated Learning

Institution: IIE Rosebank College, Pretoria CBD

Client: Community Oriented Substance Use Programme (COSUP)

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Website V1: https://tmosetlha.github.io/COSUP_WEBSITE_VERSION_1/

Table of Contents

Contents

Table of Contents.....	3
1. Introduction / Problem Domain.....	5
1.1 Background.....	5
1.2 Problem Statement.....	5
1.3 Project Scope.....	5
1.4 Organisational Context.....	6
2. Solution Domain.....	6
2.1 Functional Requirements.....	6
2.2 UML Use Case Diagram.....	8
3. Logical System Model.....	11
4. Class Diagrams.....	14
4.1 Class Diagram Table.....	14
4.2 Domain Class Diagram.....	16
5. Appendix.....	18
Appendix A — UML Diagram Note.....	18
Appendix B — References.....	18
6. Introduction.....	19
7. Logical Architectural Design — High-Level Design.....	19
7.1 Architectural Style.....	19
7.2 High-Level Architectural Description.....	21
7.3 System Input/Output Specification Reference.....	21
7.4 Functional Requirements Reference.....	21
8. Logical Architectural Design — Low-Level Design.....	22

8.1 Low-Level Design Overview	22
9. User Interaction Design.....	25
9.1 Interactions with the User — Input Interactions.....	25
9.2 Interactions with the User — Request Interactions	27
10. Database Design	29
10.1 Overview	29
Database Table 1: User	29
Database Table 2: Site	29
Database Table 3: Service.....	30
Database Table 4: Referral	31
Database Table 5: NewsItem.....	31
Database Table 6: SiteService.....	32
Database Table 7: HopelineLog.....	32
Database Table 8: Session	32
Database Table 9: AuditLog.....	33
11. Entity Relationship Diagram (ERD) Design	33
11.1 ERD Description	33
12. System Reports Design	35
13. Appendix.....	36
Appendix C — Diagram Notes (System Design).....	36
Appendix D — Technology Stack Summary.....	37
Appendix E — References	38

DOCUMENT 1 — REQUIREMENT ANALYSIS DOCUMENT

1. Introduction / Problem Domain

1.1 Background

The Community Oriented Substance Use Programme (COSUP) is a collaborative public health initiative established in 2015 through a partnership between the City of Tshwane and the University of Pretoria's Department of Family Medicine. Since its founding, COSUP has provided free, harm reduction-based services to people who use drugs (PWUD) across the Tshwane metropolitan area. Over more than a decade of continuous operation, the programme has supported in excess of 5,000 individuals through 16 active service sites.

Despite this extensive operational track record, COSUP has historically relied exclusively on physical outreach, peer networks, and word-of-mouth referrals to connect individuals with its services. The organisation has lacked any dedicated digital presence — a gap that significantly limits its ability to reach individuals who are not already embedded in existing peer networks, particularly those in geographically dispersed areas such as Mamelodi, Soshanguve, Garankuwa, and Bronkhorstspuit.

1.2 Problem Statement

The absence of a centralised digital platform creates three critical operational problems for COSUP. First, individuals who search online for substance use support services in Tshwane cannot find COSUP — the organisation is entirely absent from the most immediate and time-sensitive help-seeking channel available to people in crisis. Second, information about clinic operating hours, service availability, and programme updates is fragmented and inconsistent, relying on physical notices that may be outdated or inaccessible. Third, COSUP's reach is structurally limited to individuals who are already connected to outreach workers or peer networks, excluding the broader population of people who could benefit from harm reduction services.

This project addresses these problems by developing two integrated digital platforms: a responsive public-facing website and an Android mobile application, both designed to provide an accessible, confidential, and multilingual gateway for individuals seeking COSUP's services in Tshwane.

1.3 Project Scope

The scope of this project encompasses the full design, development, and handover of:

- A responsive website built using HTML5, CSS3, and JavaScript with a planned ASP.NET MVC / C# backend for production deployment.
- An Android mobile application developed in Kotlin, featuring a clinic locator, Hopeline crisis button, services directory, and multilingual support.
- A multilingual interface supporting seven South African languages: English, IsiZulu, Xitsonga, TshiVenda, SeTswana/Sepedi, Afrikaans, and Xhosa.
- A POPIA-compliant self-referral and user registration system.
- An administrative interface enabling authorised COSUP staff to manage clinic listings, news items, and service information.

Items explicitly excluded from scope include patient health record management, appointment booking, iOS mobile development, real-time in-app chat, and long-term hosting and maintenance.

1.4 Organisational Context

COSUP operates across 16 sites in the Tshwane metropolitan area, ranging from the main Sediba Hope Building on Bosman Street in Pretoria CBD to community sites in Soshanguve, Mamelodi, Garankuwa, Atteridgeville, Laudium, Olievenhoutbosch, Winterveldt, Hammanskraal, Bronkhorstspuit, Eersterust, and Daspoort. Services delivered include Opioid Agonist Therapy (OAT), Needle and Syringe Programmes (NSP), Psychosocial Support, Skills Development and Vocational Training, HIV and HCV Prevention, and Community Integration.

The system being developed is not a clinical system — it is a public-facing information, navigation, and communication platform designed to reduce barriers to accessing COSUP's existing services. No sensitive health data will be stored or processed.

2. Solution Domain

2.1 Functional Requirements

The following table presents the complete functional requirements of the COSUP digital platform, identifying the active actor (participant who initiates the function), the system function, and the passive actor (participant who benefits or is affected by the outcome).

Participant (Active Actor)	Function of the System	Participant (Passive Actor)
Visitor / Member of Public	View COSUP mission, history, and impact statistics	COSUP Organisation
Visitor / Member of Public	Browse full list of COSUP services and descriptions	COSUP Organisation
Visitor / Member of Public	Search and locate nearest COSUP clinic site	COSUP Clinic Staff
Visitor / Member of Public	Switch interface language (7 SA languages)	COSUP Organisation

Participant (Active Actor)	Function of the System	Participant (Passive Actor)
Person in Crisis	Tap Hopeline button to call 0800 611 197	COSUP Peer Counsellor
Person in Crisis	Tap Hopeline button to WhatsApp COSUP support line	COSUP Peer Counsellor
Prospective Client	Complete self-referral form and submit contact request	COSUP Case Worker
Prospective Client	Register an account with name, email, phone, language pref	COSUP System (Database)
Registered User	Sign in to personal COSUP account	COSUP System (Database)
Registered User	Receive email confirmation after registration	COSUP System (Email)
Mobile App User	View interactive clinic map with 16 site pins	Google Maps API
Mobile App User	View services listing on mobile	COSUP Organisation
Mobile App User	Browse news and updates feed	COSUP Communications Team
COSUP Admin Staff	Log into admin interface with role-based access	COSUP System (Database)
COSUP Admin Staff	Update clinic operating hours and addresses	Website Visitors / App Users
COSUP Admin Staff	Publish news items and announcements	Website Visitors / App Users
COSUP Admin Staff	View self-referral submissions	COSUP Case Workers
System	Send confirmation email after registration	Registered User

Participant (Active Actor)	Function of the System	Participant (Passive Actor)
System	Enforce POPIA-compliant consent on all data capture forms	All Data Subjects
System	Encrypt all data transmission via HTTPS/TLS	All Users

2.2 UML Use Case Diagram

The following use case diagram identifies the system boundary, all actors, and the use case actions of the COSUP digital platform. The system is divided into two sub-systems: the Public Website and the Android Mobile Application, both supported by a shared backend.

UML USE CASE DIAGRAM — TABLE FORMAT

System Boundary: COSUP Website & Mobile Application

ACTORS:

External Actors: Visitor/Public User | Person in Crisis | Prospective Client | Registered User
| COSUP Admin Staff

System Actors: Google Maps API | Email Service (Mailgun/SendGrid) | SQL Server Database
| POPIA Consent Module

USE CASES — Public Website Sub-system:

UC01: View Organisation Information | UC02: Browse Services | UC03: Locate Clinic Sites |
UC04: Switch Language | UC05: Access Hopeline

UC06: Submit Self-Referral Form | UC07: Register Account | UC08: Sign In | UC09: View
News & Updates

USE CASES — Mobile Application Sub-system:

UC10: Tap Hopeline (Call/WhatsApp) | UC11: View Interactive Clinic Map | UC12: Browse
Services on Mobile | UC13: View News Feed

USE CASES — Admin Sub-system:

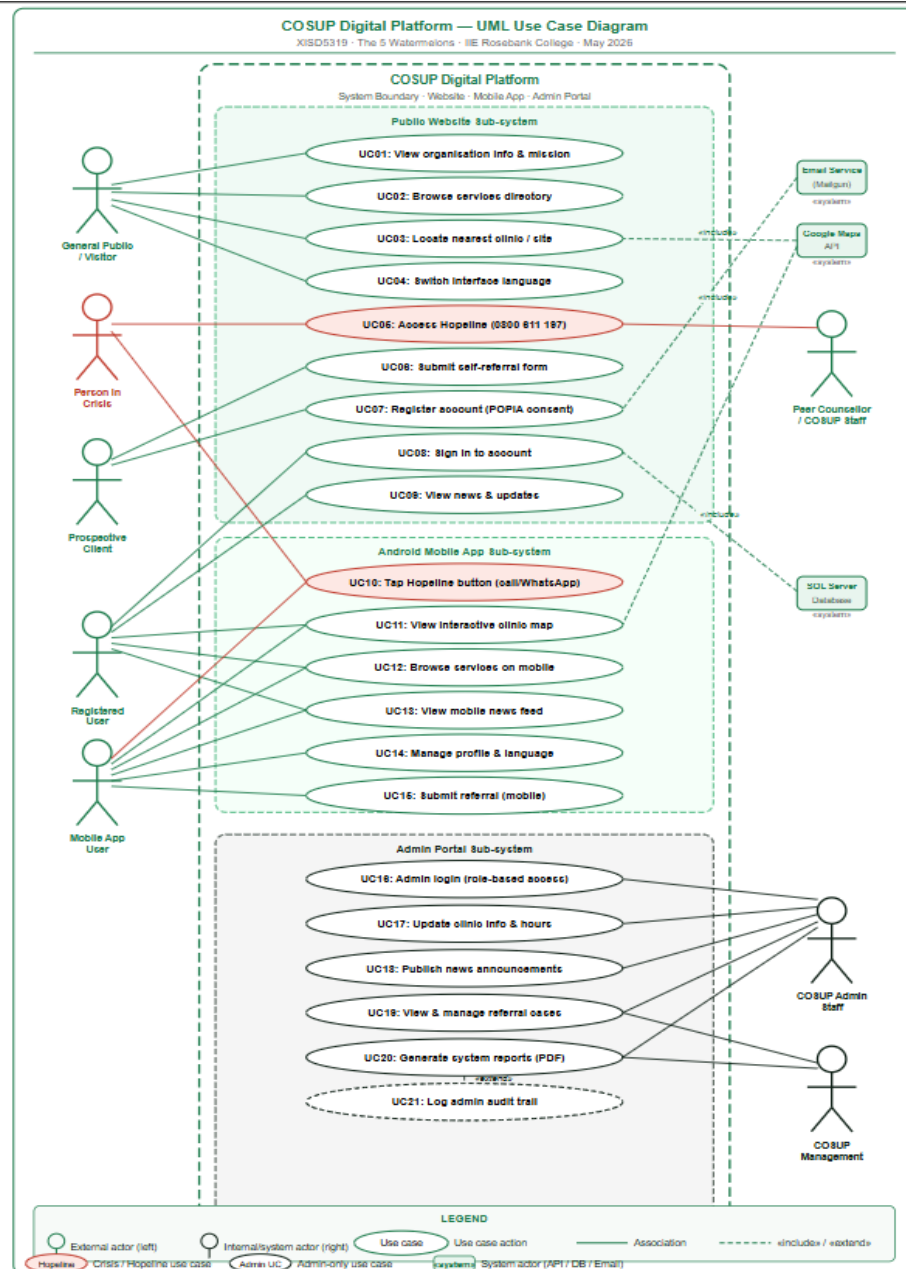
UC14: Admin Login (Role-Based) | UC15: Update Clinic Information | UC16: Publish News Items | UC17: View Referral Submissions

[Diagram tool: Draw.io — see Appendix A for full visual diagram]



THE USE CASE DIAGRAM STARTS ON THE NEXT PAGE

UML USE CASE DIAGRAM — VISUAL DIAGRAM FORMAT



3. Logical System Model

The Logical System Model defines the complete input/output specification of the COSUP platform. Each row represents a system process identified by its GUI interaction, the system method that processes it, and the entity (database table) it references. This table forms the foundation for the system design, database design, and class diagrams.

GUI Input	GUI Output	System Process (Method)	Entity Relationship (Table)
Enter name, email, phone, password, language, POPIA consent	Email confirmation sent; account created	registerUser(): validate inputs, hash password, insert User record, trigger email confirmation	User table
Enter email and password on login form	JWT token returned; user dashboard shown	authenticateUser(): verify credentials, generate JWT token, log session	User table, Session table
Click 'Find Help Near You' / search by area	Map with 16 site pins; list of clinic cards with address, hours, phone	getSites(): query all active clinic records, filter by search term, return JSON	Site table
Click clinic card / map pin	Clinic detail overlay: name, address, hours, services, phone	getSiteById(): retrieve single site record by ID	Site table, SiteService table
Click Hopeline button	Initiates phone call or WhatsApp to	triggerHopeline(): open tel: or wa.me: URI; log anonymous click event	HopelineLog table

GUI Input	GUI Output	System Process (Method)	Entity Relationship (Table)
(website or mobile app)	0800 611 197		
Enter service search / click service category	Service detail card: name, description, icon	getServices(): retrieve all services or filter by category	Service table
Click language toggle; select from 7 language options	Full UI re-renders in selected language	setLanguage(): update user language preference, load i18n resource bundle	User table (langPref column)
Enter referral form: name, contact, area, service needed	Confirmation message; case worker notified	submitReferral(): validate form, insert Referral record, send notification email	Referral table
Admin: update clinic name, address, hours, phone	Updated clinic data reflected on website and app	updateSite(): validate admin role, update Site record, log audit trail	Site table, AuditLog table
Admin: publish news article with title, body, date	News item appears in website news section and app feed	publishNews(): validate admin role, insert NewsItem record	NewsItem table
No input (system scheduled)	News feed displayed on homepage	getLatestNews(): retrieve last 5 published NewsItem records ordered by date	NewsItem table

GUI Input	GUI Output	System Process (Method)	Entity Relationship (Table)
	and mobile app		
No input (system)	All data transmissions encrypted end-to-end	enforceHTTPS(): enforce TLS certificate on all endpoints, redirect HTTP to HTTPS	System config (no DB table)
Admin: view list of referral submissions	Table of referrals with name, contact, date, area, status	getReferrals(): retrieve all Referral records, paginate, filter by status	Referral table
Mobile app: access interactive map	Live map with 16 pins, user's current location	initMap(): call Google Maps API, plot COSUP_SITES coordinate array, attach info windows	Site table (coords), Google Maps API

4. Class Diagrams

4.1 Class Diagram Table

The following table identifies each UML class derived from the Entity Relationship column in the Logical System Model. Each class has its name, properties (attributes with data types and constraints), and related classes. For a domain class diagram, only names, attributes, and relationships are shown — no operations.

Name of Entity (UML Class)	Properties of Entity (UML Class)	Related To
User	userId: INT (PK, auto-increment) firstName: VARCHAR(50) NOT NULL lastName: VARCHAR(50) NOT NULL email: VARCHAR(100) UNIQUE NOT NULL password: VARCHAR(255) NOT NULL (hashed) phone: VARCHAR(20) langPref: VARCHAR(10) DEFAULT 'en' emailVerified: BOOLEAN DEFAULT FALSE createdAt: DATETIME	Session, Referral
Session	sessionId: INT (PK) userId: INT (FK → User) jwtToken: VARCHAR(500) createdAt: DATETIME expiresAt: DATETIME	User
Site	siteId: INT (PK) siteName: VARCHAR(100) NOT NULL address: VARCHAR(200) NOT NULL latitude: DECIMAL(9,6) longitude: DECIMAL(9,6) phone: VARCHAR(30) operatingHours: VARCHAR(100) isMainCentre: BOOLEAN isActive: BOOLEAN DEFAULT TRUE	SiteService
SiteService	siteServiceId: INT (PK)	Site, Service

Name of Entity (UML Class)	Properties of Entity (UML Class)	Related To
	siteId: INT (FK → Site) serviceId: INT (FK → Service)	
Service	serviceId: INT (PK) serviceName: VARCHAR(100) NOT NULL serviceDescription: TEXT iconPath: VARCHAR(200) category: VARCHAR(50)	SiteService
Referral	referralId: INT (PK) userId: INT (FK → User, nullable) fullName: VARCHAR(100) NOT NULL contactNumber: VARCHAR(30) preferredArea: VARCHAR(100) serviceNeeded: VARCHAR(100) consentGiven: BOOLEAN NOT NULL status: VARCHAR(30) DEFAULT 'pending' submittedAt: DATETIME	User
NewsItem	newsId: INT (PK) title: VARCHAR(200) NOT NULL body: TEXT NOT NULL publishedAt: DATETIME author: VARCHAR(100) isPublished: BOOLEAN DEFAULT FALSE	—
HopelineLog	logId: INT (PK) channelType: VARCHAR(20) — 'call' or 'whatsapp' clickedAt: DATETIME platform: VARCHAR(20) — 'web' or 'mobile'	—
AuditLog	logId: INT (PK) adminUserId: INT (FK → User) action: VARCHAR(200) targetTable: VARCHAR(50) targetId: INT performedAt: DATETIME	User

4.2 Domain Class Diagram

The domain class diagram below models the relationships between all identified classes. Associations are indicated by directional arrows, generalisations by inheritance lines, and aggregations where one class logically contains another.

DOMAIN CLASS DIAGRAM — COSUP SYSTEM

ASSOCIATIONS:

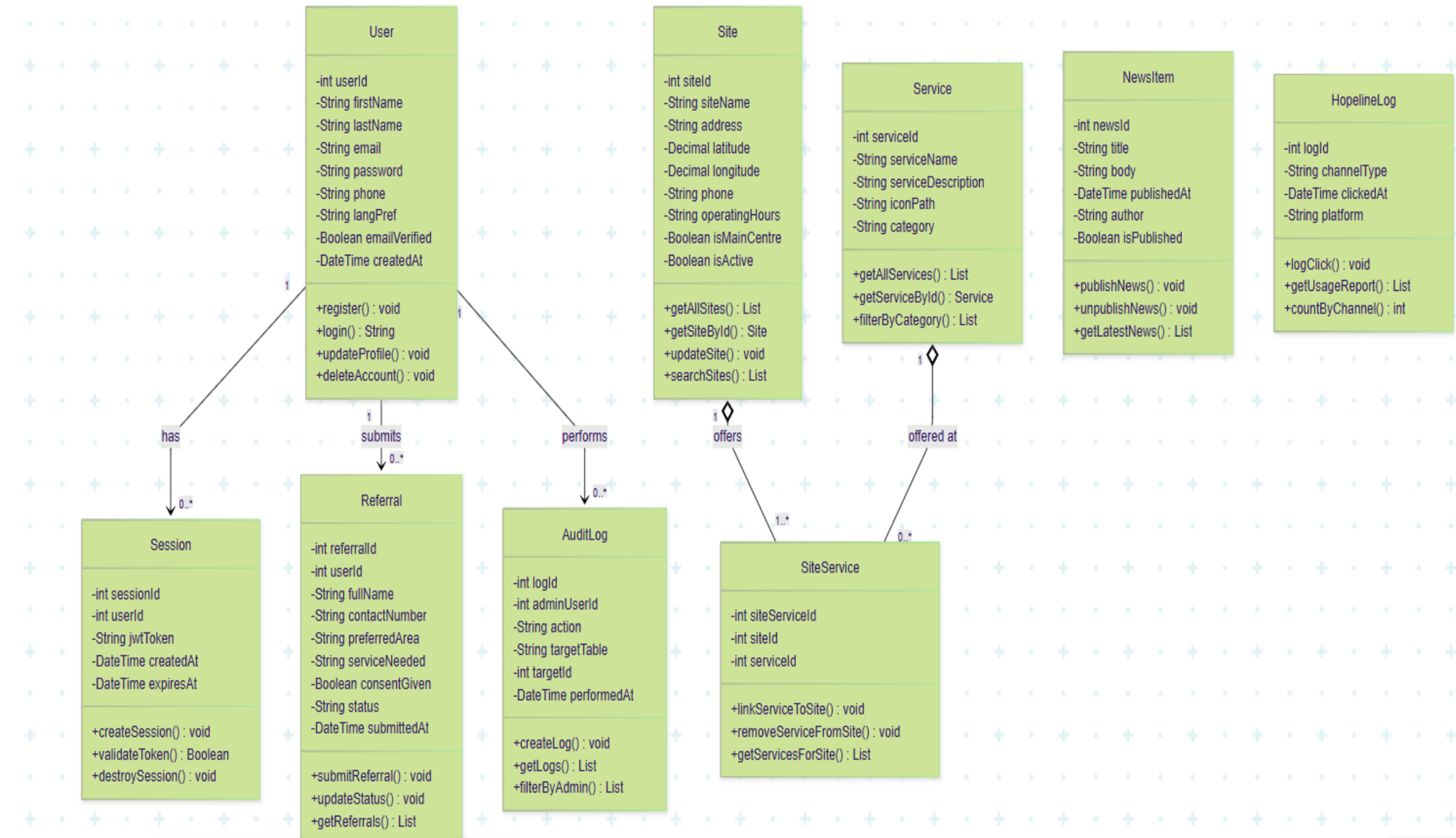
- User (1) ————— (0..*) Session [One user can have many sessions]
- User (1) ————— (0..*) Referral [One user can submit many referrals]
- User (1) ————— (0..*) AuditLog [Admin user creates many audit log entries]

AGGREGATIONS (Site contains services via junction):

- Site (1) ————— (1..*) SiteService [One site offers one or more services]
- Service (1) ————— (0..*) SiteService [One service can be offered at many sites]

INDEPENDENT ENTITIES (no direct associations):

- NewsItem — published independently by admin staff
 - HopelineLog — records anonymous crisis button events (no user FK)
- [Full UML class diagram generated in Draw.io — see Appendix A]

DOMAIN CLASS DIAGRAM – VISUAL FORMAT

5. Appendix

Appendix A — UML Diagram Note

The UML Use Case Diagram and Domain Class Diagram described in Sections 2.2 and 4.2 respectively were designed in Draw.io (<https://draw.io>) following the specifications detailed in this document. The complete visual diagram files are available in the project's GitHub repository at https://tmosetlha.github.io/COSUP_WEBSITE_VERSION_1/ and are submitted alongside this document as separate visual artefacts.

Appendix B — References

Ambler, S.W., 2004. The object primer: Agile model-driven development with UML 2.0. 3rd ed. Cambridge: Cambridge University Press.

Dennis, A., Wixom, B.H. and Roth, R.M., 2018. Systems analysis and design. 6th ed. Hoboken: John Wiley & Sons.

Department of Justice and Constitutional Development, 2013. Protection of Personal Information Act, No. 4 of 2013. Pretoria: Government Printer.

Fowler, M., 2003. UML distilled: A brief guide to the standard object modelling language. 3rd ed. Boston: Addison-Wesley.

IIE Rosebank College, 2026. XISD5319 Module Manual: Work Integrated Learning. Pretoria: IIE Rosebank College.

Pressman, R.S. and Maxim, B.R., 2019. Software engineering: A practitioner's approach. 9th ed. New York: McGraw-Hill Education.

The 5 Watermelons, 2026. COSUP Website & Mobile Application — Project Charter. Pretoria: IIE Rosebank College.

DOCUMENT 2 — SYSTEM DESIGN DOCUMENT

6. Introduction

This System Design Document translates the requirements specified in the Requirement Analysis Document (Document 1) into concrete technical design specifications for the COSUP digital platform. The document covers the logical architectural design at both high and low levels, the user interaction design including input and request interactions, the database table design with example data, the Entity Relationship Diagram, the system reports design, and a full reference list.

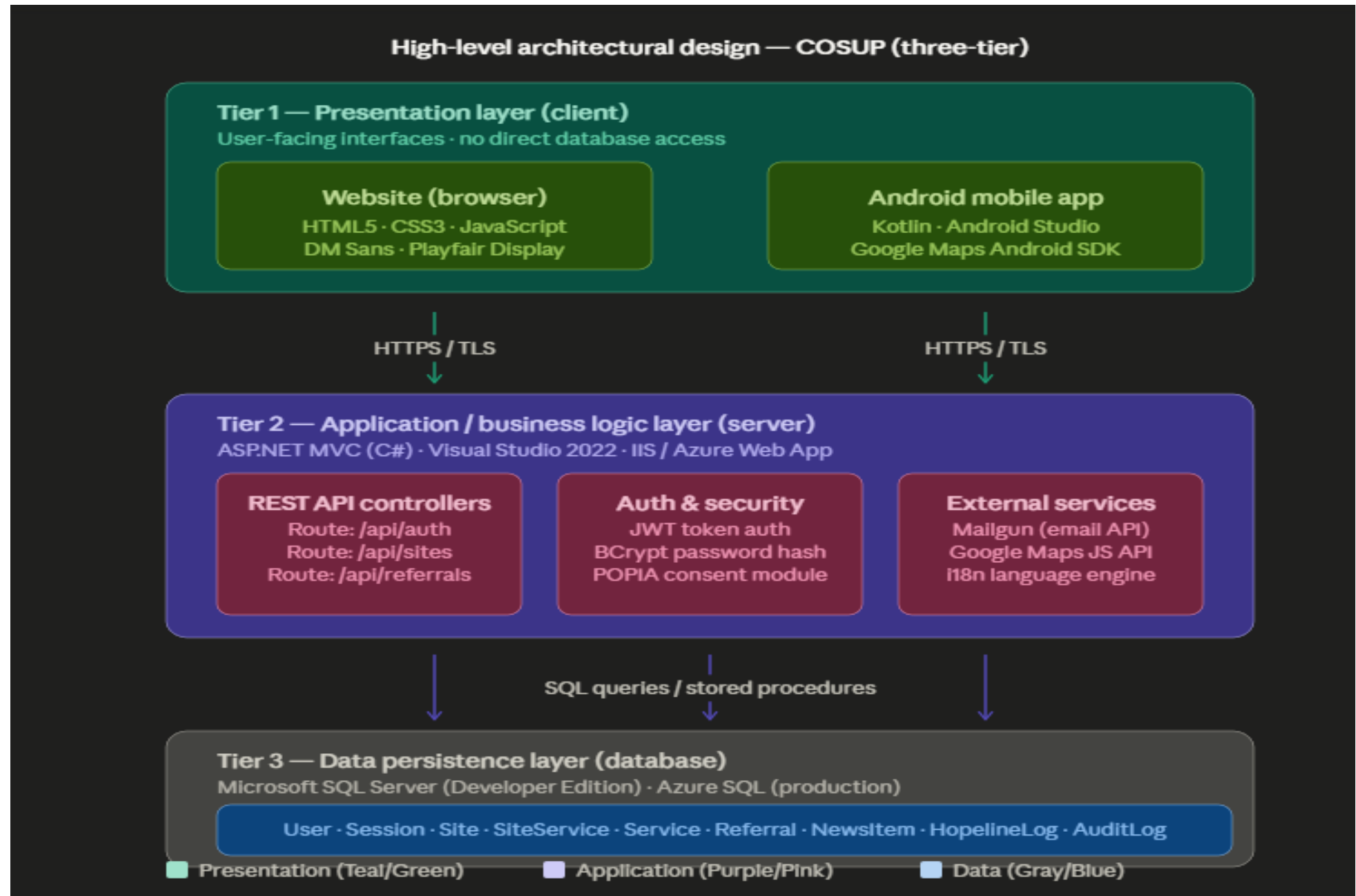
The system being designed consists of two client-facing platforms — a responsive website and an Android mobile application — supported by a shared ASP.NET MVC / C# backend and a Microsoft SQL Server database. The design follows a three-tier architectural model in line with industry best practice for web-based systems (Pressman and Maxim, 2019).

All design decisions are grounded in the functional requirements established in Section 2 of Document 1 and are directly traceable to the client needs of COSUP as documented in the Project Charter (The 5 Watermelons, 2026).

7. Logical Architectural Design — High-Level Design

7.1 Architectural Style

The COSUP system is designed as a three-tier (three-level) client-server architecture. This architectural style separates the system into three distinct logical layers — the Presentation Tier, the Application/Logic Tier, and the Data Tier — each with clearly defined responsibilities and interfaces.



7.2 High-Level Architectural Description

The COSUP system operates as follows at a high level. Clients (website browsers and Android mobile app users) connect to the Application Server over HTTPS (TLS encrypted). The Application Server processes all business logic — authentication, data retrieval, form submission handling, language selection, and admin operations — and queries or writes to the SQL Server database as required. The Google Maps API is called from the client side (JavaScript / Kotlin) to render the interactive clinic map, using coordinate data supplied by the Application Server.

Functional building blocks are allocated as follows. All read operations (site listings, service listings, news feed, clinic search) are handled by the Application Server and cached where appropriate. All write operations (user registration, login, referral submission, admin updates) are processed exclusively by the Application Server before any database interaction occurs. No direct client-to-database communication exists — all data access is mediated through the application layer. The email confirmation system is handled by an external email service provider (Mailgun or SendGrid) called via API from the Application Server.

7.3 System Input/Output Specification Reference

For the complete Input/Output specification, refer to the Logical System Model Table in Section 3 of Document 1. That table maps each user interaction to its corresponding system process and entity table, providing the full I/O specification for all 14 identified system processes.

7.4 Functional Requirements Reference

For the complete functional requirements table, refer to Section 2.1 of Document 1. That table identifies all 20 functional requirements with active actors, system functions, and passive actors, forming the basis for the architectural allocation decisions described above.

8. Logical Architectural Design — Low-Level Design

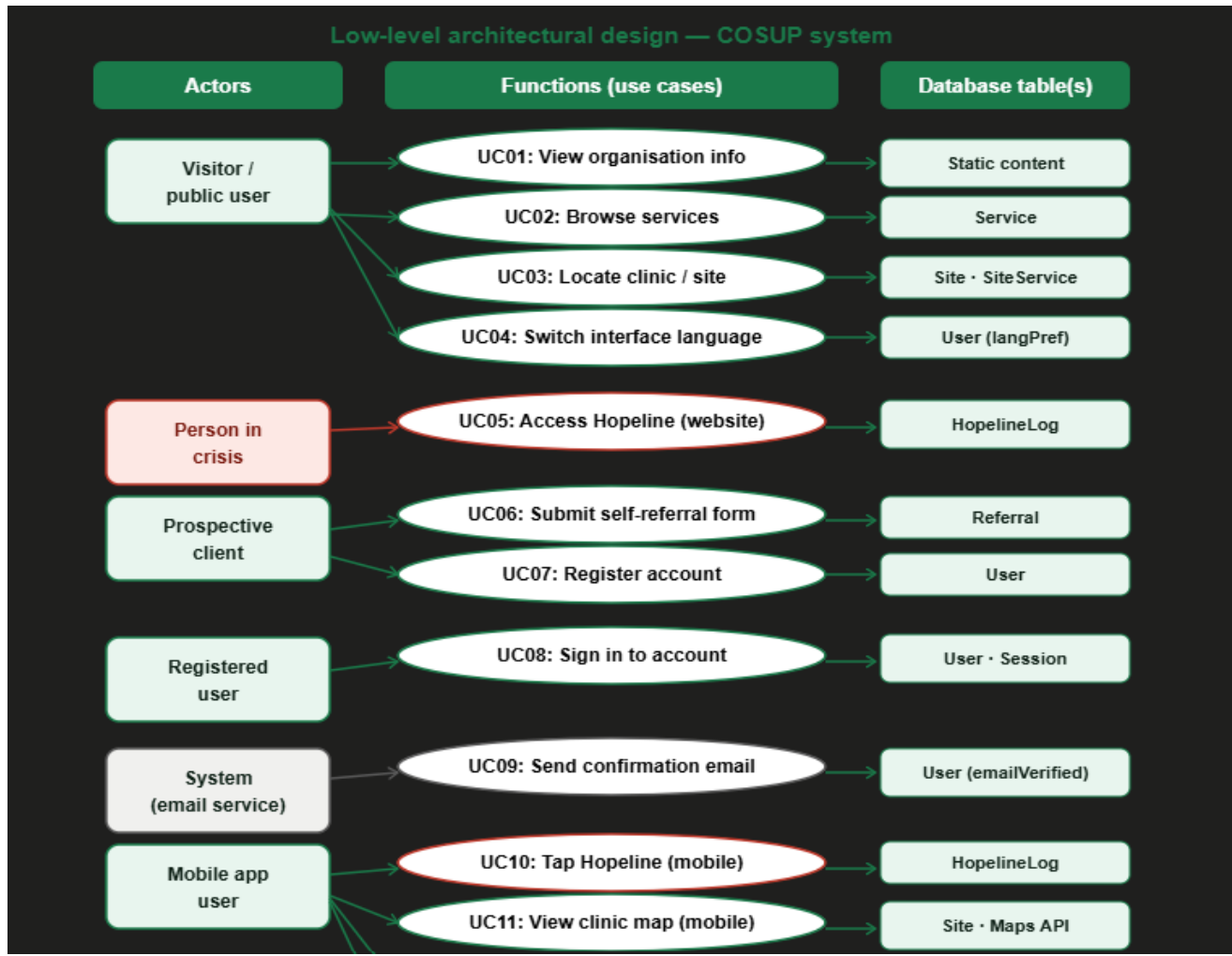
8.1 Low-Level Design Overview

The low-level design diagram models the relationships between actors, use cases (functions), and the database tables they interact with. It consists of three columns: Actors (from the use case diagrams in Section 2.2), Functions (the system use cases), and the Database tables each function references.

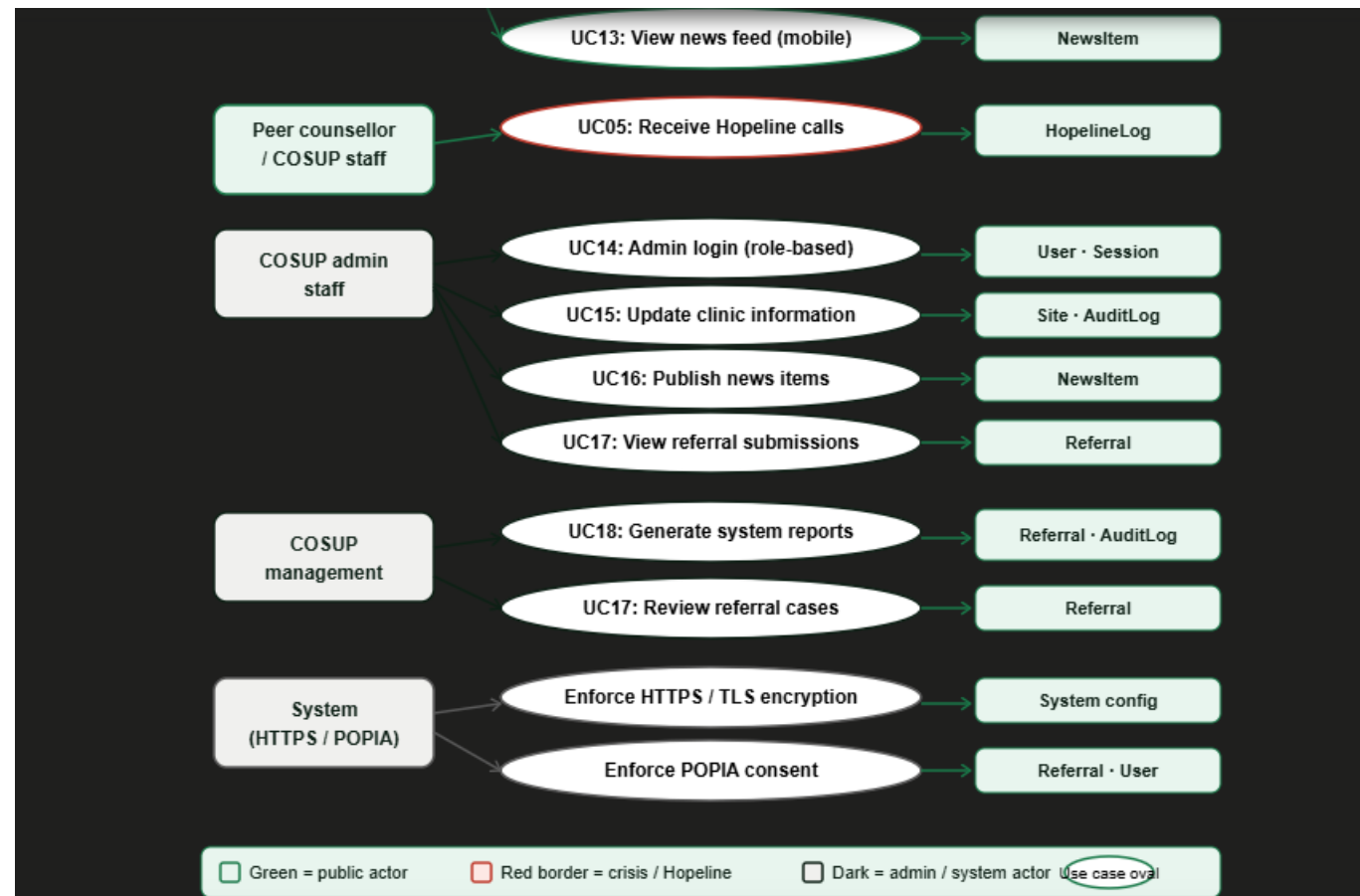


THE DIAGRAM STARTS ON THE NEXT PAGE, ITS ONE DIAGRAM, YET DIVIDED INTO TWO PARTS FOR VISUAL EMPHASIS!!!

First Part of Low-Level Design



Second Part of Low-Level Design



Note: Arrows in the full Draw.io low-level diagram connect each actor to their use cases, and each use case to its database table(s), following the format shown in the IIE Module Manual (IIE Rosebank College, 2026, p. 7). The diagram is included in Appendix C.

9. User Interaction Design

9.1 Interactions with the User — Input Interactions

Input interactions represent data controls used to read data into the system. These are the first reference to the GUI containing the complete interaction specification. The following table presents a structured annotated list of all input menus, indicating the hierarchical relationship between interaction levels.

Menu Level	Input Category	Input Field / Interaction	Description
1st Level	Registration Form	First Name	Text input — VARCHAR(50), required
1st Level	Registration Form	Last Name	Text input — VARCHAR(50), required
1st Level	Registration Form	Email Address	Email input — VARCHAR(100), required, must be unique
1st Level	Registration Form	Phone Number	Tel input — VARCHAR(30), optional
1st Level	Registration Form	Password	Password input — min 8 chars, hashed before storage
2nd Level	Registration Form	Preferred Language	Select dropdown — 7 SA languages
2nd Level	Registration Form	POPIA Consent Checkbox	Checkbox — required; must be ticked to submit
1st Level	Login Form	Email Address	Email input — validated against User table
1st Level	Login Form	Password	Password input — verified against hashed value
1st Level	Self-Referral Form	Full Name	Text input — VARCHAR(100), required

Menu Level	Input Category	Input Field / Interaction	Description
1st Level	Self-Referral Form	Contact Number	Tel input — VARCHAR(30)
1st Level	Self-Referral Form	Preferred Area	Text input — matches COSUP site areas
2nd Level	Self-Referral Form	Service Needed	Dropdown — OAT, NSP, Psychosocial, Skills Dev, HIV/HCV, Community
2nd Level	Self-Referral Form	POPIA Consent	Checkbox — required
1st Level	Clinic Search	Search Query	Text input — filters Site table by name or area
1st Level	Language Toggle	Language Selection	Button group — sets langPref; 7 options
1st Level	Admin Login	Admin Email	Email input — validated against User table (role = admin)
1st Level	Admin Login	Admin Password	Password input — verified against hashed value
2nd Level	Admin: Update Clinic	Clinic Name, Address, Hours, Phone	Text inputs — updates Site table record
2nd Level	Admin: Publish News	Title, Body, Publish Date	Text / textarea inputs — inserts NewsItem record

9.2 Interactions with the User — Request Interactions

Request interactions represent all service requests to the system and include requests for functional processing and outputs in the form of screen displays (graphical representations). These refer to menus and output parameter forms used to specify services provided by the system.

Request Type	User Action / Trigger	System Response / Output	Notes
Screen Display	User navigates to Home page	Hero section, stats bar (10+ years, 5,000+ helped, 16 sites), service cards, Hopeline box	Static content + dynamic stats from DB
Screen Display	User navigates to 'Find a Site'	Interactive Google Maps with 16 pins; clinic card grid with name, address, hours, phone number	Site table → Google Maps API integration
Screen Display	User navigates to 'Our Services'	6 service cards: OAT, NSP, Psychosocial, Skills Dev, HIV/HCV, Community Integration	Service table
Screen Display	User navigates to 'About'	Organisation story, partnership cards (City of Tshwane, UP), impact stats	Static + DB stats
Screen Display	User navigates to 'Contact'	3 contact option cards + Hopeline box with 0800 611 197	Static content
Screen Display	User opens Registration modal	Registration form with all fields, language dropdown, POPIA consent	Frontend form, POST to /api/auth/register
Screen Display	User opens Login modal	Login form with email and password fields	Frontend form, POST to /api/auth/login
Screen Display	Successful registration	Email confirmation modal; confirmation email sent to user	Email Service API (Mailgun/SendGrid)
Screen Display	Admin views referrals dashboard	Table of referral submissions with columns: name, contact, area, service, date, status	Referral table, role-based access

Request Type	User Action / Trigger	System Response / Output	Notes
Printed / PDF Report	Admin exports referrals report	PDF report of all referral submissions for a given period	Referral table → PDF generation
Printed / PDF Report	Admin exports site listing	PDF of all 16 COSUP site details for distribution	Site table → PDF generation
Mobile Screen Display	App user opens Home screen	Welcome screen with Hopeline button, 4 service tiles, impact stats, news feed	API call → Service, NewsItem tables
Mobile Screen Display	App user opens 'Find Help'	Interactive Kotlin map with 16 site pins; list of all sites below	Site table, Google Maps Android SDK
Mobile Screen Display	App user taps Hopeline button	Confirmation dialog → launches phone dialler or WhatsApp with 0800 611 197	HopelineLog table

10. Database Design

10.1 Overview

The COSUP database consists of nine tables derived from the Class Diagram Table in Section 4.1. Each table is presented below with its primary key, secondary/foreign keys, data fields, and three example rows demonstrating realistic data. All tables are designed to Third Normal Form (3NF) as required by the module specification.

Database Table 1: User

PK userID	firstName	lastName	email	password (hash)	phone	langPref	emailVerified	createdAt
1	Tshiamo	Mosetlha	tshiamo@email.com	\$2b\$12\$abc...	072 123 4567	en	TRUE	2026-05-01
2	Zanele	Dlamini	zanele@gmail.com	\$2b\$12\$xyz...	083 987 6543	zu	TRUE	2026-05-02
3	Thabo	Nkosi	thabo.n@outlook.com	\$2b\$12\$mno...	071 555 0011	tn	FALSE	2026-05-05

Database Table 2: Site

PK siteId	siteName	address	latitude	longitude	phone	operatingHours	isMain	isActive
1	Sediba Hope — Bosman St	173 Bosman St, Pretoria CBD	- 25.7450	28.1880	082 858 4304	Mon–Fri 08:00–16:00	TRUE	TRUE
2	COSUP Bronkhorstspuit	Zithobeni Clinic, Bronkhorstspuit	- 25.7990	28.7420	076 560 1389	Tue & Thur 09:30–14:00	FALSE	TRUE

PK siteId	siteName	address	latitude	longitude	phone	operatingHours	isMain	isActive
3	COSUP Garankuwa	Reatlegile Centre, Stand 05030	- 25.6000	28.0200	079 321 4152	Mon–Fri 08:00– 16:00	FALSE	TRUE

Database Table 3: Service

PK serviceId	serviceName	serviceDescription	iconPath	category
1	Opioid Agonist Therapy	Evidence-based medication-assisted treatment to reduce dependence and support long-term recovery.	IMAGES/opiod.png	Treatment
2	Needle & Syringe Programme	Free sterile equipment to reduce the risk of HIV, Hepatitis C and blood-borne infections.	IMAGES/needle.png	Prevention
3	Psychosocial Support	Counselling, peer support groups and mental health services for people who use drugs.	IMAGES/psych.png	Support

Database Table 4: Referral

PK referralId	FK userId	fullName	contactNumber	preferredArea	serviceNeeded	consentGiven	status	submittedAt
1	NULL	Anonymous User	072 000 0001	Soshanguve	Opioid Agonist Therapy	TRUE	pending	2026-05-03
2	2	Zanele Dlamini	083 987 6543	Mamelodi	Psychosocial Support	TRUE	contacted	2026-05-04
3	NULL	Keamogetswe M.	061 222 3344	Garankuwa	Needle & Syringe	TRUE	pending	2026-05-07

Database Table 5: NewsItem

PK newsId	title	body	publishedAt	author	isPublished
1	16 Sites Now Active Across Tshwane	COSUP now operates 16 community sites across the Tshwane metro. Full coverage from Bronkhorstspuit to Garankuwa.	2026-03-15	COSUP Comms	TRUE
2	Peer Support Groups — Now Weekly on Thursdays	Weekly peer support sessions are now available every Thursday across all main sites in Tshwane.	2026-04-01	COSUP Comms	TRUE
3	COSUP Celebrates 10+ Years of Harm Reduction	2026 marks over a decade of free, compassionate, evidence-based harm reduction services for Tshwane.	2026-05-01	COSUP Comms	TRUE

Database Table 6: SiteService

PK siteServiceId	FK siteId (→ Site)	FK serviceId (→ Service)
1	1 (Sediba Hope)	1 (Opioid Agonist Therapy)
2	1 (Sediba Hope)	2 (Needle & Syringe Programme)
3	2 (Bronkhorstspuit)	2 (Needle & Syringe Programme)

Database Table 7: HopelineLog

PK logId	channelType	clickedAt	platform
1	call	2026-05-03 14:22:01	web
2	whatsapp	2026-05-04 09:11:45	mobile
3	call	2026-05-07 21:03:18	mobile

Database Table 8: Session

PK sessionId	FK userId	jwtToken	createdAt	expiresAt
1	1	eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...	2026-05-01 08:00	2026-05-01 20:00
2	2	eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...	2026-05-02 10:30	2026-05-02 22:30
3	1	eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...	2026-05-05 14:00	2026-05-06 02:00

Database Table 9: AuditLog

PK logId	FK adminUserId	action	targetTable	targetId	performedAt
1	1	Updated operating hours for Site 2	Site	2	2026-05-03 09:15
2	1	Published NewsItem: '16 Sites Now Active'	NewsItem	1	2026-05-05 11:00
3	1	Updated referral status to 'contacted' for Referral 2	Referral	2	2026-05-07 14:22

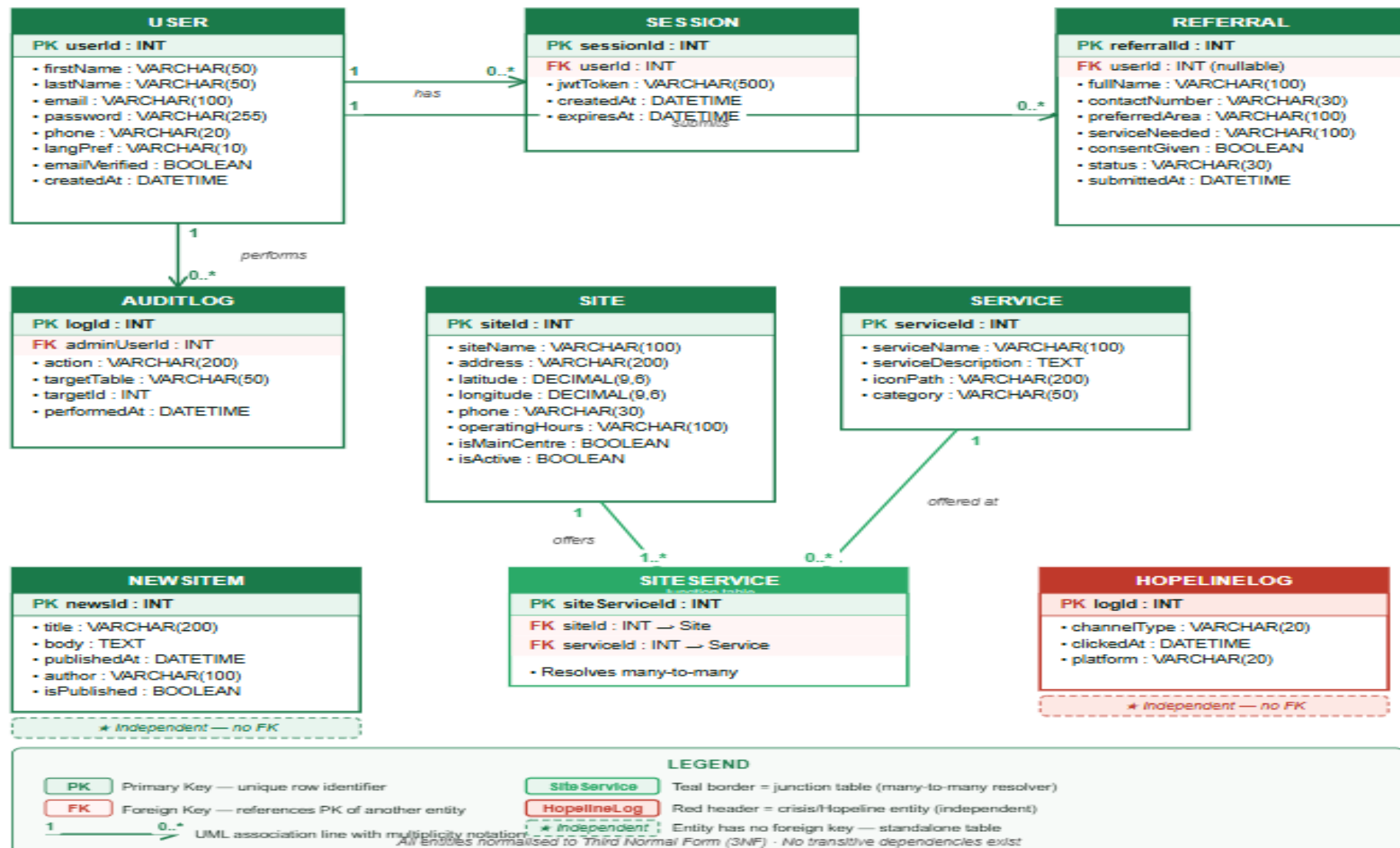
11. Entity Relationship Diagram (ERD) Design

11.1 ERD Description

The COSUP database ERD models nine entities and their relationships. The following describes the cardinality and nature of each relationship. All relationships are in Third Normal Form (3NF) — no transitive dependencies exist.

COSUP DATABASE — ENTITY RELATIONSHIP DIAGRAM

UML Notation · XISD5319 · The 5 Watermelons · IIE Rosebank College



COSUP Website & Mobile Application · XISD5319 Work Integrated Learning · The 5 Watermelons · IIE Rosebank College · May 2026

Github: https://tmosetlha.github.io/COSUP_WEBSITE_VERSION_1/

ERD DIAGRAM — COSUP DATABASE (Crow's Foot Notation)

User — (1 : 0..*) —> Session
 User — (1 : 0..*) —> Referral
 User — (1 : 0..*) —> AuditLog
 Site — (1 : 1..*) —> SiteService <— (1 : 0..*) — Service
 NewsItem [independent — no FK]
 HopelineLog [independent — no FK]

[Full ERD drawn in Draw.io using Crow's Foot notation — see Appendix C]

12. System Reports Design

As part of the database and ERD design, the following reports have been identified as required by COSUP. Each report specifies the data source, trigger, and output format.

Report Name	Trigger / Who Requests	Data Source (Tables)	Output Format	Key Fields Displayed
Referral Submissions Report	COSUP Admin Staff — on demand or scheduled	Referral table, User table	PDF download / Screen table	Full Name, Contact, Area, Service Needed, Consent, Status, Submission Date
Site Listing Report	COSUP Admin / Public — on demand	Site table, SiteService table, Service table	PDF download / Printable web page	Site Name, Address, Phone, Hours, Services Offered
Hopeline Usage Report	COSUP Management — monthly	HopelineLog table	Screen chart + CSV export	Channel Type (call/WhatsApp), Platform, Date/Time, Count per Day

Report Name	Trigger / Who Requests	Data Source (Tables)	Output Format	Key Fields Displayed
Registered Users Report	COSUP Admin — on demand	User table	Screen table (admin panel)	Name, Email, Phone, Language Preference, Email Verified, Registration Date
Admin Activity Audit Report	COSUP IT / Management — on demand	AuditLog table, User table	Screen table + PDF download	Admin Name, Action, Target Table/ID, Date/Time of Action
Services Directory	Public — always visible on website and app	Service table	Screen display (website + mobile)	Service Name, Description, Icon, Category

13. Appendix

Appendix C — Diagram Notes (System Design)

The following visual diagrams are required to accompany this document and are submitted separately as Draw.io files:

- Low-Level Architectural Design Diagram (Section 8) — actors connected to use cases connected to database tables, following the three-column format prescribed in the IIE Module Manual.
- ERD Diagram (Section 11) — all nine entities with their attributes, primary keys, foreign keys, and Crow's Foot notation cardinality lines drawn in Draw.io.
- All diagrams are also hosted on the project GitHub: https://tmosetlha.github.io/COSUP_WEBSITE_VERSION_1/

Appendix D — Technology Stack Summary

Layer	Technology	Purpose
Frontend (Website)	HTML5, CSS3, JavaScript (DM Sans + Playfair Display fonts)	Responsive public-facing website
Frontend (Mobile)	Kotlin, Android Studio	Android mobile application
Backend	ASP.NET MVC, C#, Visual Studio 2022	REST API, business logic, authentication
Database	Microsoft SQL Server (Developer Edition)	All entity data storage
Authentication	JWT (JSON Web Token), BCrypt password hashing	Secure user session management
Email	Mailgun / SendGrid API	Registration confirmation emails
Maps	Google Maps API (website JS) + Google Maps Android SDK	Interactive clinic locator
Hosting	Azure Web App + Azure SQL (production)	Live deployment environment
Compliance	POPIA Act 4 of 2013	Data protection and privacy

Appendix E — References

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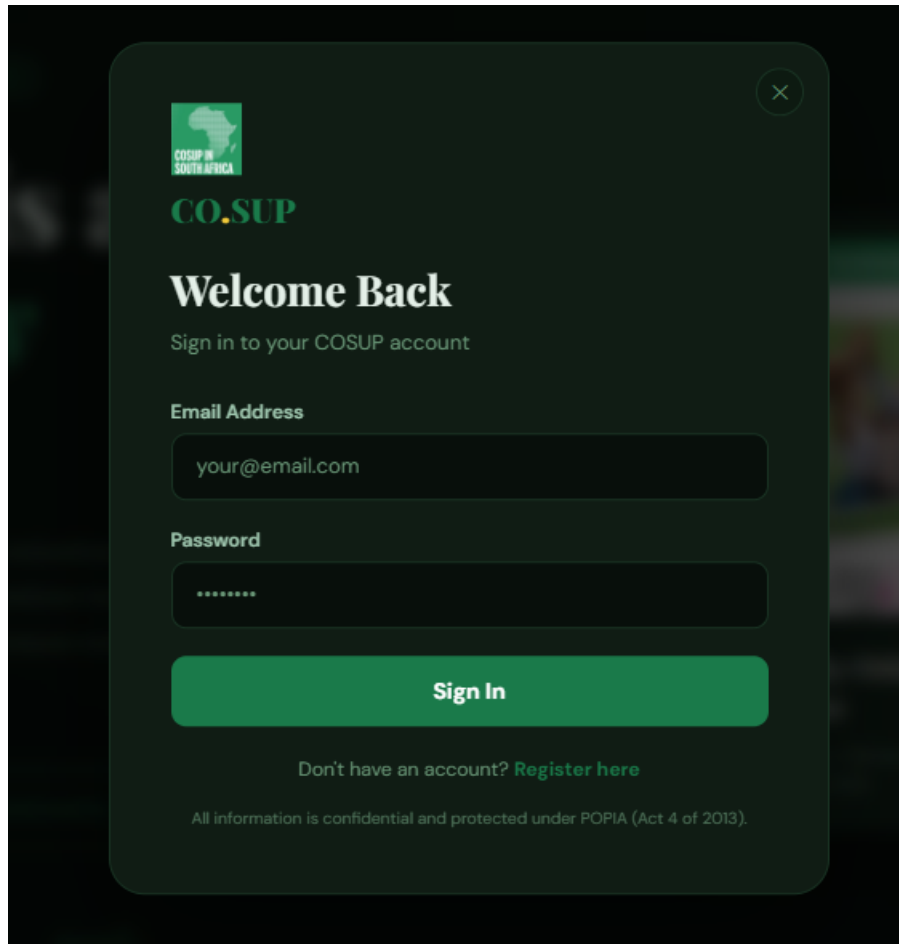
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The 5 Watermelons, 2026. COSUP Website Version 1 [Online]. Available at: https://tmosetlha.github.io/COSUP_WEBSITE_VERSION_1/ [Accessed: 9 May 2026].

The 5 Watermelons, 2026. COSUP UI/UX Prototype [Online]. Available at: <https://tmosetlha.github.io/cosup-prototype/> [Accessed: 9 May 2026].

SCREENSHOTS FOR COSUP WEBSITE AND MOBILE APPLICATION

SIGN IN PAGE



The screenshot displays a dark-themed sign-in modal for COSUP. At the top left is the COSUP logo, which includes a map of South Africa and the text 'COSUP IN SOUTH AFRICA' and 'CO.SUP'. A close button (X) is in the top right corner. The main heading is 'Welcome Back' in white, followed by the subtitle 'Sign in to your COSUP account'. Below this are two input fields: 'Email Address' with the placeholder 'your@email.com' and 'Password' with masked characters '.....'. A large green 'Sign In' button is positioned below the password field. At the bottom, there is a link 'Don't have an account? Register here' and a small disclaimer: 'All information is confidential and protected under POPIA (Act 4 of 2013)'.

REGISTER PAGE

 **CO.SUP**

Create Account

Join the COSUP community — it's free and confidential

First Name
First name

Last Name
Last name

Email Address
your@email.com

Phone Number
+27 XX XXX XXXX

Password
Min 8 characters

Preferred Language
English

☐ I consent to COSUP contacting me and understand my data is protected under POPIA.

Create Account & Send Confirmation

Already have an account? [Sign in here](#)

A confirmation email will be sent to verify your address. All data is encrypted and protected under POPIA (Act 4 of 2013).

WEBSITE – DARK MODE HOMEPAGE

The screenshot displays the COSUP website homepage in dark mode. The header features the COSUP logo, navigation links (Home, Services, Find a Site, About, Contact), and user options (Sign In, Register) alongside a helpline number (0800 611 197). The main content area includes a location tag for Tshwane, South Africa, with 16 active sites. A large headline reads 'Recovery is a community journey.' Below this, a paragraph describes the free harm reduction services provided. Two buttons, 'Find Help Near You' and 'Join Our Community', are positioned at the bottom of the main section. On the right, a featured section titled 'Community Oriented Substance Use Programme' highlights the partnership between the City of Tshwane and the University of Pretoria, noting their pioneering work since 2015. This section includes a graphic with the text 'THE MODEL' and 'DRUG AND SUBSTANCE USE', and a badge indicating they are 'Nationally recognised'.

COSUP

Home Services Find a Site About Contact

Sign In Register **905 Helpline: 0800 611 197**

TSHWANE, SOUTH AFRICA · 16 ACTIVE SITES

Recovery is a *community* journey.

COSUP provides free, compassionate harm reduction services to people who use drugs in Tshwane. We combine healthcare, human rights and community support — because everyone deserves a path forward.

Find Help Near You **Join Our Community**

Services open now

EST. 2015 · TSHWANE

COSUP

THE MODEL

DRUG AND SUBSTANCE USE **#Helpline**

Community Oriented Substance Use Programme

City of Tshwane × University of Pretoria — pioneering harm reduction since 2015.

Nationally recognised

LIGHT MODE - HOMEPAGE

[Home](#) [Services](#) [Find a Site](#) [About](#) [Contact](#)

[Sign In](#) [Register](#) [Hopeline: 0800 611 197](#)

TSHWANE, SOUTH AFRICA · 16 ACTIVE SITES

Recovery is a *community* journey.

COSUP provides free, compassionate harm reduction services to people who use drugs in Tshwane. We combine healthcare, human rights and community support — because everyone deserves a path forward.

[Find Help Near You](#) [Join Our Community](#)

Services open now

EST. 2015 · TSHWANE



THE MODEL



DRUG AND SUBSTANCE USE  #hopeline



Community Oriented Substance Use Programme

City of Tshwane x University of Pretoria — pioneering harm reduction since 2015.

 **Nationally recognised**

SERVICES PAGE

WHAT WE OFFER

Comprehensive care for every step of your journey

All services are free of charge. No referral needed. Walk in or contact us directly.



Opioid Agonist Therapy

Evidence-based medication-assisted treatment to reduce dependence and support long-term recovery in a safe, non-judgemental environment.



Needle & Syringe Programme

Free, sterile equipment to reduce the risk of HIV, Hepatitis C and other blood-borne infections for people who inject drugs.



Psychosocial Support

Counselling, peer support groups and mental health services to address the emotional and social aspects of substance use.



Skills Development & Vocational Training

Skills development and employment support to help clients rebuild their lives and contribute meaningfully to their communities.



HIV & HCV Prevention

Testing, treatment linkage and education programmes to prevent and manage HIV and Hepatitis C among people who use drugs.



Community Integration

Peer-led initiatives that embed services within existing public health infrastructure, reducing stigma and increasing access.

FIND A SITE PAGE

⚠️ Powered by Google Maps API — Replace YOUR_API_KEY in index.html to activate

🔍 Search by area, site name or service...

01 COSUP Bronkhorstspuit 📍 Zithobeni Clinic, Bronkhorstspuit 🕒 Tue & Thur · 09:30–14:00 ☎️ 076 560 1389	02 COSUP Hammanskraal 📍 Mandisa Shiceka Clinic, Old Warmbaths Road 🕒 Mon & Fri · 08:00–16:00 ☎️ 066 300 8338	03 COSUP Sediba Hope Building Main Centre 📍 173 Bosman Street, Pretoria 🕒 Mon–Fri · 08:00–16:00 ☎️ 082 858 4304	04 COSUP Daspoort Poli Clinic 📍 Cnr Market & Camp Street, Pretoria West 🕒 Mon–Fri · 08:00–16:00 ☎️ 082 857 0922
05 COSUP Atteridgeville Clinic 📍 1A Mareka Street, Atteridgeville 🕒 Mon–Fri · 08:00–16:00 ☎️ 082 858 2450	06 COSUP Laudium Community 📍 405 Bengal Street, Laudium 🕒 Mon–Fri · 08:00–16:00 ☎️ 081 725 2462	07 COSUP Olievenhoutbosch 📍 Cnr Imbovane & Imbongolo Street 🕒 Mon–Fri · 08:00–16:00 ☎️ 066 472 5740	08 COSUP Winterveldt 📍 1/4 Cemetery Road, Slovo Gardens 🕒 Mon–Fri · 08:00–16:00 ☎️ 071 573 5883
09 COSUP Mamelodi Lusaka 📍 Cnr Millenyane & Ratshwene Street 🕒 Mon–Fri · 08:00–16:00 ☎️ 082 858 2553 / 072 592 6754	10 COSUP Mamelodi Ikageng 📍 21882 Molokoloko Circle, Mamelodi (Holy Bible Church) 🕒 Mon–Fri · 08:00–16:00 ☎️ 076 560 1389	11 COSUP Mamelodi Regional Hospital 📍 Cnr Serapeng & Tsarnaya Road, Mamelodi 🕒 Mon–Fri · 08:00–16:00 ☎️ 066 472 5580	12 COSUP Eersterust Community Health Centre 📍 Cnr PS Fourie Drive & Hans Coverdale Road 🕒 Mon–Fri · 08:00–16:00 ☎️ 082 941 6038
13 COSUP Soshanguve Block V 📍 Elim Tabernacle Church, Inkanyezi Street 🕒 Mon–Thur · 08:00–16:00 ☎️ 066 300 8338	14 COSUP Soshanguve Block K POPUP 📍 Cnr Aubrey Matlala & Tlou Street 🕒 Mon–Fri · 08:00–16:00 ☎️ 076 587 6770	15 COSUP Soshanguve M17 📍 Dream Team Foundation, M17 Road, Soshanguve South 🕒 Mon–Fri · 08:00–16:00 ☎️ 082 858 2563	16 COSUP Garankuwa 📍 Reatlegile Centre, Stand 05030 🕒 Mon–Fri · 08:00–16:00 ☎️ 079 321 4152

ABOUT US PAGE

OUR STORY

A decade of compassionate, evidence-based care

COSUP was established in response to the growing limitations of abstinence-based and punitive approaches to substance use in Tshwane. We believe that people who use drugs deserve dignified, rights-based healthcare.

Through a partnership between the City of Tshwane and the University of Pretoria's Department of Family Medicine, COSUP has become a nationally and internationally recognised model for harm reduction.

[Get Involved](#)



City of Tshwane

Primary funder and operational partner



University of Pretoria

Department of Family Medicine · Research partner

CONTACT US PAGE

NEED SUPPORT?

We're here for you — no judgement, no barriers

All COSUP services are free of charge. You do not need a referral. Just reach out.



Find a Clinic

16 sites across Tshwane — find the one nearest to you using our map above.

[View All Sites](#)



Self-Referral

Register and submit a referral request — a COSUP team member will contact you within 24 hours.

[Register Now](#)



Email Us

Research and programme enquiries.

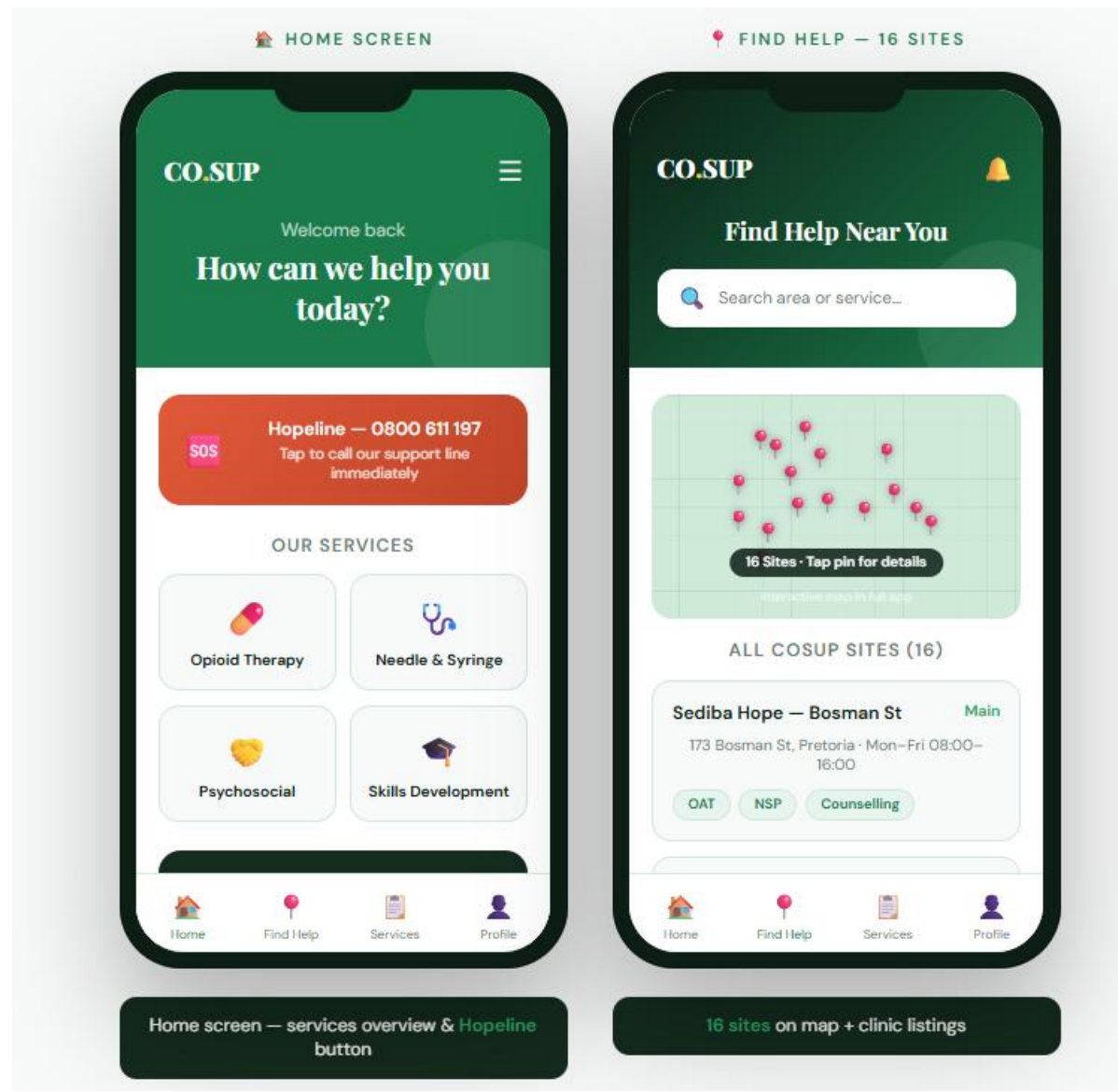
up.ac.za/up-copc-research-unit

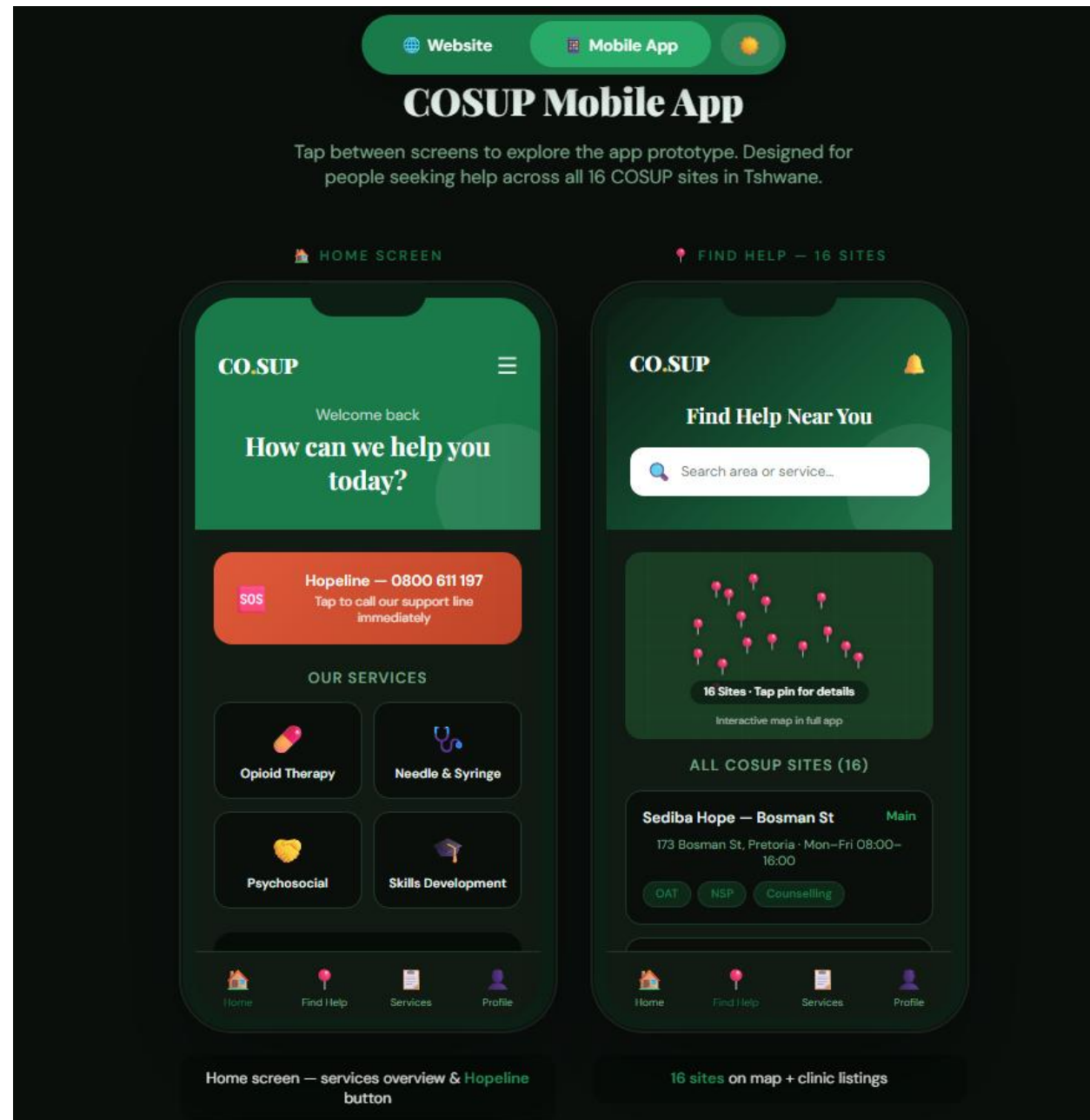
Hopeline

Available 7 days a week · Confidential · Free · No judgement

0800 611 197

MOBILE APPLICATION - LIGHT MODE





SEE YALL SOON!

